

Hans-Peter Marshall

Cryosphere Geophysics And Remote Sensing (CryoGARS) group
Department of Geosciences and CGISS, Boise State University
Environmental Research Building (ERB) 1104 (lab), 3155 (office)
1910 University Drive
Boise, ID 83725

(208) 426-1416 (office)

(303) 859-3106 (cell)

hpmarshall@boisestate.edu

<https://www.boisestate.edu/earth-cryogars/>

Education

- 2005 Ph.D, Civil Engineering (Geotechnical), GPA: 3.9/4.0
Institute of Arctic and Alpine Research, University of Colorado at Boulder
Thesis: *Snowpack spatial variability: towards understanding its effect on remote sensing measurements and snow slope stability*
- 1999 B.S., Physics with honors, Geophysics Minor (1st in UW history), GPA: 3.6/4.0
University of Washington
Thesis: *Wet snow densification*

Selected Professional Experience

- 08/2018-8/2021 Project Scientist, NASA SnowEx Mission
- 08/2022-present Professor, Department of Geosciences, Boise State University
- 06/2022-10/2023 Co-organizer of the 2024 International Snow Science Workshop (avalanche science)
- 01/2015-present Wilderness First Responder, Avalanche Level 1 & 2

Selected Recent Peer-Reviewed Publications

[h-index=33, i10-index=70, citations=3336]

Marshall, H.P., Deeb, E., Forster, R., Vuyovich, C., Elder, K., Hiemstra, C., and Lund, J. (2021). L-band InSAR depth retrieval during the NASA SnowEx 2020 Campaign: Grand Mesa, Colorado. *Proceedings of the International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 625–627, DOI:10.1109/IGARSS47720.2021.9553852

Arenson, L., Colgan, W., and **Marshall, H.P.** (2021, 2nd edition). *Physical, Thermal, and Mechanical Properties of Snow, Ice, and Permafrost*, chapter 2, pages 35–75. Snow and ice-related hazards, risks, and disasters. Elsevier, ed. Haeberli and Whiteman, Radarweg 29, PO Box 211, 1000 AE Amsterdam, Netherlands

Marshall, H.P., Schweizer, J., and Birkeland, K. (2011). Some recent advances in snow and avalanche science. *Cold Regions Science and Technology*, 69(2-3):119–121

Marshall, H.P. and Johnson, J. B. (2009). Accurate inversion of high-resolution snow penetrometer signals for microstructural and micromechanical properties. *Journal of Geophysical Research-Earth Surface*, 114(F04016)

Marshall, H.P. and Koh, G. (2008). FMCW radars for snow research. *Cold Regions Science and Technology*, 52(2):118–131